ELEC5305 Project Proposal Guide

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1 Project Title

Give your project a clear and descriptive title.

Example: "Real-Time Noise Suppression for Mobile Devices Using Deep Learning"

2 Student Information

- Full Name
- Student ID (SID)
- GitHub Username
- GitHub Project Link: `https://github.com/username/elec5305-project-[SID]`

3 Project Overview

Briefly describe the goal of your project. Include:

- The problem you are addressing
- Why it is important
- Your proposed solution

Example: This project aims to develop a real-time noise suppression algorithm for mobile devices using a lightweight neural network. The goal is to improve speech intelligibility in noisy environments.

4 Background and Motivation

Provide context and motivation for your project.

- What existing solutions or research are relevant?
- Why did you choose this topic?

Example: Recent advances in deep learning have enabled effective noise suppression, but many models are too large for mobile deployment. This project explores efficient architectures suitable for embedded systems.

5 Proposed Methodology

Outline your approach:

- Tools and platforms (e.g., MATLAB, Python)
- Signal processing techniques (e.g., short-time Fourier transform (STFT), Wiener filtering, neural networks)
- Data sources (e.g., TIMIT, DNS Challenge dataset)

6 Expected Outcomes

Describe what you expect to deliver:

- A working prototype or simulation
- Performance metrics (e.g., signal-to-noise ratio (SNR) improvement, Perceptual Evaluation of Speech Quality (PESQ) score)
- GitHub documentation and demonstration

7 Timeline (Weeks 1–13)

Week	Task
1–2	Consider Project Topic
3–5	Literature review and dataset collection
6–9	Initial implementation and testing
10–11	Optimisation and evaluation
12–13	Final report and GitHub documentation

8 References

Include at least three academic or technical peer-reviewed references.

Example:

1. Loizou, P. C. (2013). *Speech Enhancement: Theory and Practice.*
2. Reddy, C. et al. (2021). DNS Challenge: Improving Noise Suppression Models.
3. Ephraim, Y., & Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator.

9 Appendix (Optional)

Include any diagrams, preliminary results, or additional notes.

10 Rubric

- **Clarity and Structure (20%)** – Proposal is well-organised and clearly written.
- **Feasibility (20%)** – Project scope is achievable within the timeline.
- **Technical Depth (20%)** – Demonstrates understanding of signal processing concepts.
- **References (20%)** – Includes at least three peer-reviewed papers.
- **GitHub Integration (20%)** – Proposal is uploaded to GitHub and the link is provided.

11 Word Limit

Your proposal should be between 700 and 1000 words. Be concise and focused.

12 GitHub Setup Instructions

To complete your submission, you must create a GitHub account and host your proposal in a public repository.

Step-by-Step Instructions

1. Go to `https://github.com` and click **Sign Up**.
2. Create a username and password, and verify your email address.
3. Once logged in, click the + icon in the top-right corner and select **New repository**.
4. Name your repository `elec5305-project-[Your-SID]` and choose **Public**.
5. Click **Create repository**.
6. Upload your proposal PDF using the **uploading an existing file** link.
7. Create a `README.md` file briefly describing your project.
8. Enable GitHub Pages:
   - Go to **Settings** → **Pages**.
   - Under **Source**, select the `main` branch and click **Save**.
   - Your site will be published at:

     `https://yourusername.github.io/elec5305-project-[Your-SID]`
9. Include this GitHub Pages link in your proposal PDF.